

ARUN MALLIKARJUNA

Full-Stack & AI Platform Engineer | Custom Agents · LLM Integration · System Design

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PROFESSIONAL SUMMARY

Full-stack engineer with 4+ years building production systems end to end and 2+ years integrating LLM APIs into core product features. I build custom AI agents and write the system prompts behind them, run evals, and ship internal AI tooling — while owning architecture, design patterns, reliability, and scaling. Strong on React and Node.js, REST APIs, SQL and NoSQL data modeling, async backends, and thorough QA. I work directly from PRDs and customer problems to ship features that actually solve them.

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Tecnotree Convergence

Jan 2024 – Present · Bengaluru, India

- Owned the architecture and design — patterns, reliability, and scaling — of a rule-based Promotion Engine, serving as the team's go-to for technical decisions and letting business teams launch telecom campaigns without redeployment.
- Led incident response when systems failed: debugged and fixed production issues across distributed microservices, sustaining 99.5% availability under high transaction volume.
- Ran thorough QA — SIT, smoke, and UAT — and reviewed teammates' code against requirements and PRDs before release.
- Built REST APIs, reporting dashboards, and third-party integrations, aligning delivery to clear business and customer outcomes.
- Improved frontend performance ~30% via optimized React rendering and state management.

Software Engineer — Tecnotree Convergence

Oct 2021 – Dec 2023 · Bengaluru, India

- Built scalable, high-performance REST APIs in Node.js for high-volume transaction processing, using async concurrency to keep latency low under load.
- Contributed to the microservices architecture and design, improving end-to-end throughput and reliability.
- Translated a real customer problem into a shipped feature — configurable workflows that let non-technical teams self-serve, cutting turnaround time.
- Optimized rule-evaluation logic to reduce per-request latency and improve efficiency.

PROJECTS

Tailorwright — AI Agent Platform

Claude API · Next.js · TypeScript · Node.js

- Built custom AI agents and authored the system prompts behind them — a Claude Code skill, a subagent, and a CLI on one shared core — for agentic résumé workflows.
- Integrated LLM APIs with structured prompt orchestration and a no-fabrication guardrail, and ran evals against a deterministic scoring core.
- Shipped it as internal-style AI tooling with automated tests, SSRF hardening, and input limits for reliability and security.

Free Resume Pro — AI SaaS Platform

Next.js · React · Node.js · Gemini API · Vercel

- Built a core product feature around the Gemini API — ATS scoring, job-description matching, and AI rewriting — using structured prompt engineering.
- Implemented user authentication (JWT) and a privacy-first, zero-storage design; owned the full lifecycle end to end as the sole developer and scaled it on Vercel.

Promotion Manager System — Internal Platform

Node.js · React · Microservices · REST APIs

- Designed and built a scalable promotion-management system on a microservices backend, with a real-time rule engine enabling non-technical teams to configure campaigns without deployments.

Algorithm Visualization Platform — Personal Project

JavaScript · React

- Built interactive, step-by-step visualizations of core algorithms and data structures — demonstrating strong DSA and complexity-analysis intuition.

TECHNICAL SKILLS

AI & Agents

LLM API Integration, Custom AI Agents, System Prompt Engineering, RAG, Vector Search, Evals, Claude API, Gemini API

Languages & Frontend

JavaScript (ES6+), TypeScript, Python · React, React Native, Next.js, Tailwind CSS

Backend & APIs

Node.js, NestJS, Express.js, REST APIs, async/await & event loop, Django, Flask

Databases

PostgreSQL, MongoDB, DynamoDB, MongoDB Atlas (schema design)

Architecture & CS

Microservices, System Design, Design Patterns, Reliability & Scaling, Data Structures, Algorithms, Complexity Analysis

Auth, Security & QA

Firebase Auth, JWT, Clerk, VAPT, SIT, Smoke, UAT, Automated Testing

Cloud & DevOps

AWS, GCP, CI/CD, Jenkins, Vercel, Railway

Methodologies

Agile, Scrum

EDUCATION

Bachelor of Engineering — Computer Science

Maharaja Institute of Technology, Mysuru · Graduated 2021